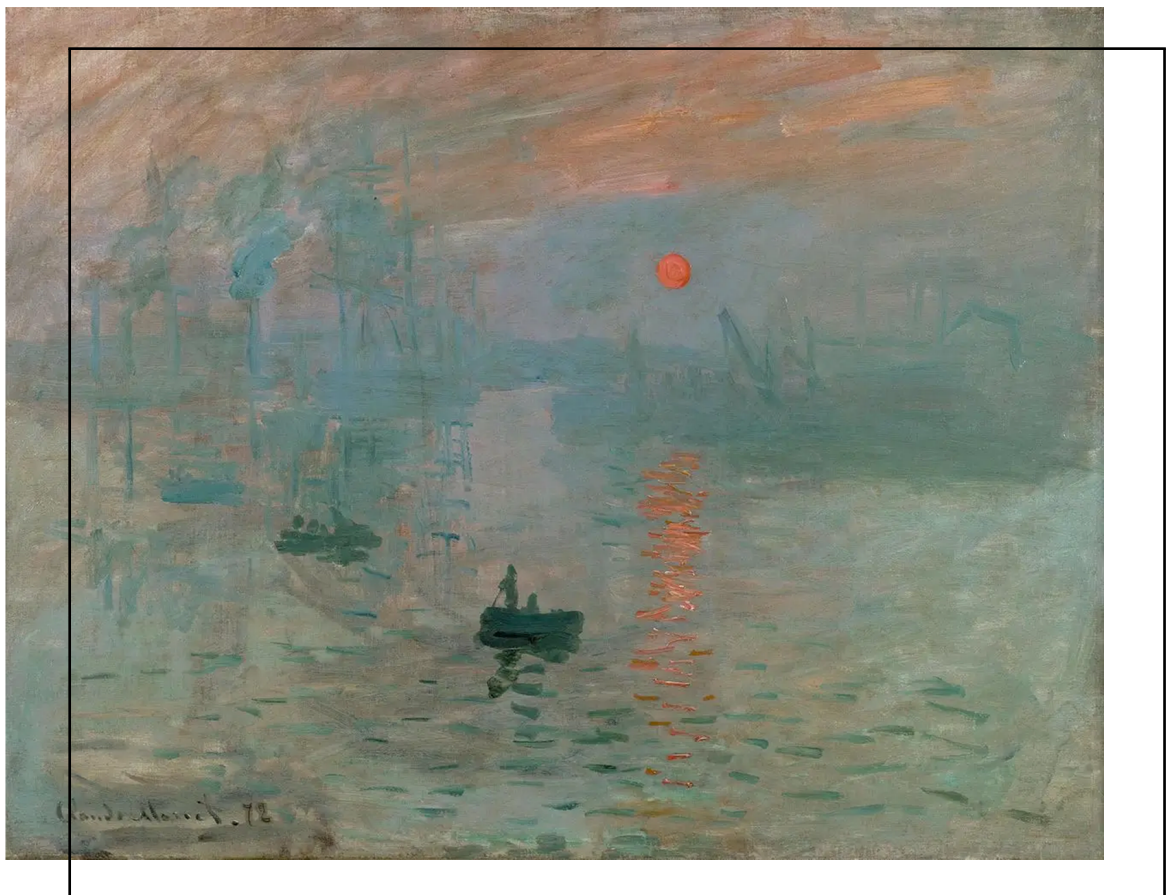


01

Lista de Exercícios

Relatividade Geral - A Teoria da Gravitação de Einstein

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Q. 01

Gravitational Field: a sphere, pt. 1

Find the gravitational field inside and outside a sphere of radius R , which carries a uniform volume mass density ρ .

To find the gravitational field inside the sphere, we can use Gauss's law, which states that the gravitational flux through a closed surface is proportional to the enclosed mass. So, we can consider a spherical Gaussian surface of radius r (where $r < R$) centered at the center of the sphere. The mass enclosed by this Gaussian surface is given by:

$$M_{\text{enclosed}} = \int_V \rho(\mathbf{r}') d^3\mathbf{r}' = \rho \int_0^r \int_0^\pi \int_0^{2\pi} (r')^2 \sin \theta' d\phi' d\theta' dr' = \frac{4}{3}\pi r^3 \rho$$

The gravitational field at a distance r from the center of the sphere can be found using Gauss's law too:

$$\oint_S \mathbf{g}(\mathbf{r}') \cdot d^2\mathbf{r}' = -4\pi G M_{\text{enclosed}}$$

Since the gravitational field is radial and has the same magnitude at every point on the Gaussian surface, we can simplify this to:

$$g(r)4\pi r^2 = -4\pi G \left(\frac{4}{3}\pi r^3 \rho \right)$$

This gives us the gravitational field inside the sphere:

$$\mathbf{g}(\mathbf{r}) = -\frac{4}{3}\pi G \rho r \hat{\mathbf{r}}, \quad \text{for } r < R \quad (1.1)$$

To find the gravitational field outside the sphere, we can use the same argument, but now the Gaussian surface is a sphere of radius r (where $r > R$) that encloses the entire mass of the sphere. Therefore, the enclosed mass is:

$$M_{\text{enclosed}} = \frac{4}{3}\pi R^3 \rho$$

Using the same application of Gauss's law to find the gravitational field, we have:

$$g(r)4\pi r^2 = -4\pi G M_{\text{enclosed}} = -4\pi G \left(\frac{4}{3}\pi R^3 \rho \right)$$



This gives us the gravitational field outside the sphere:

$$\mathbf{g}(\mathbf{r}) = -\frac{4}{3}\pi G\rho\frac{R^3}{r^2}\hat{\mathbf{r}}, \quad \text{for } r > R \quad (1.2)$$



Q. 02 Gravitational Field: a spherical shell, pt. 1

Find the gravitational field inside and outside a spherical shell of radius R , which carries a uniform surface mass density σ .

The gravitational field inside the spherical shell is zero, since there is no mass enclosed by any Gaussian surface inside the shell. Therefore, we have:

$$\mathbf{g}(\mathbf{r}) = \mathbf{0}, \quad \text{for } r < R \quad (2.1)$$

To find the gravitational field outside the spherical shell, we can use Gauss's law again. The mass enclosed by a Gaussian surface of radius r (where $r > R$) is given by:

$$M_{\text{enclosed}} = \int_S \sigma(\mathbf{r}') d^2\mathbf{r}' = \sigma \int_0^\pi \int_0^{2\pi} R^2 \sin \theta' d\phi' d\theta' = 4\pi R^2 \sigma$$

We can use again that:

$$\oint_S \mathbf{g}(\mathbf{r}') \cdot d^2\mathbf{r}' = -4\pi G M_{\text{enclosed}}$$

Since the gravitational field is radial and has the same magnitude at every point on the Gaussian surface, we can simplify this to:

$$g(r)4\pi r^2 = -4\pi G M_{\text{enclosed}} = -4\pi G (4\pi R^2 \sigma)$$

This gives us the gravitational field outside the spherical shell:

$$\mathbf{g}(\mathbf{r}) = -\frac{4\pi G R^2 \sigma}{r^2} \hat{\mathbf{r}}, \quad \text{for } r > R \quad (2.2)$$



Q. 03

Gravitational potential: a straight wire

Find the gravitational potential a distance s from an infinitely long straight wire that carries a uniform line charge λ . Compute the gradient of your potential and find the corresponding gravitational field.

To find the gravitational potential at a distance s from an infinitely long straight wire, we can use:

$$U(\mathbf{s}) = G \int \frac{dm}{\|\mathbf{s}\|}$$

Since the wire is infinitely long, we can consider a small segment of the wire of length dz , which has a mass $dm = \lambda dz$. The distance from this segment to the point where we want to calculate the potential is given by $r = \sqrt{s^2 + z^2}$. Taking a segment with length $2L$ centered at $z = 0$, we can write:

$$U(s) = G \int_{-L}^L \frac{\lambda}{\sqrt{s^2 + z^2}} dz = G\lambda \int_{-L}^L \frac{1}{\sqrt{s^2 + z^2}} dz$$

The indefinite integral has a known solution:

$$\int \frac{1}{\sqrt{s^2 + z^2}} dz = \ln(z + \sqrt{s^2 + z^2}) + C$$

So, the potential can be calculated as:

$$U(s) = G\lambda \left[\ln(L + \sqrt{s^2 + L^2}) - \ln(-L + \sqrt{s^2 + L^2}) \right] = G\lambda \ln \left(\frac{L + \sqrt{s^2 + L^2}}{-L + \sqrt{s^2 + L^2}} \right)$$

The argument of the logarithm can be simplified as:

$$\frac{L + \sqrt{s^2 + L^2}}{-L + \sqrt{s^2 + L^2}} = \frac{L + \sqrt{s^2 + L^2}}{-L + \sqrt{s^2 + L^2}} \cdot \frac{L + \sqrt{s^2 + L^2}}{L + \sqrt{s^2 + L^2}} = \frac{(L + \sqrt{s^2 + L^2})^2}{s^2}$$

For large L , we can approximate $\sqrt{s^2 + L^2} \approx L$, which gives us:

$$U(s) = 2G\lambda \ln \left(\frac{2L}{s} \right) = 2G\lambda \ln(2L) - 2G\lambda \ln(s)$$

The first term diverges as $L \rightarrow \infty$, but does not depend on s , so it has no physical effect to the gravitational field. This can be absorbed into a reference potential, i.e., the difference:

$$U(s) - U(s_0) = -2G\lambda \ln(s) + 2G\lambda \ln(s_0) = -2G\lambda \ln \left(\frac{s}{s_0} \right)$$



Choosing s_0 such that $U(s_0) = 0$, we can write the potential as:

$$U(s) = -2G\lambda \ln\left(\frac{s}{s_0}\right) \quad (3.1)$$

To find the gravitational field, we can compute the gradient of the potential:

$$\mathbf{g}(\mathbf{s}) = \nabla U(s) = -2G\lambda \nabla \left[\ln\left(\frac{s}{s_0}\right) \right] = -2G\lambda \frac{d}{ds} \left[\ln\left(\frac{s}{s_0}\right) \right] \hat{\mathbf{s}} = -2G\lambda \frac{1}{s} \hat{\mathbf{s}}$$

Concluding that:

$$\mathbf{g}(\mathbf{s}) = -\frac{2G\lambda}{s} \hat{\mathbf{s}} \quad (3.2)$$



Q. 04 Gravitational potential: a hollow sphere

The gravitational potential $U(\theta)$ is specified on the surface of a hollow sphere of radius R . Find the potential inside the sphere.

Knowing that $\nabla \cdot \mathbf{g} = -4\pi G\rho$, we can use that $\mathbf{g} = \nabla U$ to obtain the Poisson equation for the gravitational potential $\nabla^2 U = -4\pi G\rho$. In the case of a hollow sphere, we have $\rho = 0$ everywhere, so the Poisson equation reduces to the Laplace equation $\nabla^2 U = 0$. The general solution to the Laplace equation in spherical coordinates can be written as:

$$U(r, \theta) = \sum_{\ell=0}^{\infty} \left[A_{\ell} r^{\ell} + \frac{B_{\ell}}{r^{\ell+1}} \right] P_{\ell}(\cos \theta)$$

Inside the sphere, the solution must be finite at $r = 0$, which implies that $B_{\ell} = 0$ for all ℓ . Therefore, the potential inside the sphere can be written as:

$$U_{\text{inside}}(r, \theta) = \sum_{\ell=0}^{\infty} A_{\ell} r^{\ell} P_{\ell}(\cos \theta)$$

On the surface of the sphere, we have $U(R, \theta) = U(\theta)$, which gives us:

$$U(\theta) = \sum_{\ell=0}^{\infty} A_{\ell} R^{\ell} P_{\ell}(\cos \theta)$$

Multiplying both sides by $P_{\ell'}(\cos \theta)$ and integrating over $d \cos \theta$ from -1 to 1 :

$$\int_{-1}^1 U(\theta) P_{\ell'}(\cos \theta) d \cos \theta = \sum_{\ell=0}^{\infty} A_{\ell} R^{\ell} \int_{-1}^1 P_{\ell}(\cos \theta) P_{\ell'}(\cos \theta) d \cos \theta$$

Using the orthogonality of the Legendre polynomials, we have for the integral on the right-hand side the result $\frac{2}{2\ell'+1} \delta_{\ell\ell'}$, therefore:

$$\int_{-1}^1 U(\theta) P_{\ell'}(\cos \theta) d \cos \theta = A_{\ell'} R^{\ell'} \frac{2}{2\ell' + 1}$$

implying that:

$$A_{\ell} = \frac{2\ell + 1}{2R^{\ell}} \int_{-1}^1 U(\theta) P_{\ell}(\cos \theta) d \cos \theta$$



Thus, the potential inside the sphere can be expressed as:

$$U_{\text{inside}}(r, \theta) = \sum_{\ell=0}^{\infty} \frac{2\ell+1}{2R^\ell} r^\ell P_\ell(\cos \theta) \int_{-1}^1 U(\theta) P_\ell(\cos \theta) d \cos \theta \quad (4.1)$$



Q. 05

Gravitational potential: a sphere, pt. 2

The potential at the surface of a sphere of radius R is given by $U(\theta) = k \cos(3\theta)$, where k is a constant. Find the potential inside and outside the sphere, as well as the surface mass density $\sigma(\theta)$ on the sphere (assume there is no mass inside or outside the sphere).



Q. 06 Gravitational potential: a spherical shell, pt. 2

Derive the exact gravitational potential for a spherical shell of radius R which carries a surface mass density $\sigma(\theta) = k \cos(\theta)$, where k is a constant. Calculate also the quadrupole moment of this mass distribution.
